

Project Proposal and Methodology

Project Scope

This project implements a controlled customer-support generation system using three progressively enhanced systems: baseline intent generation, SOP-assisted retrieval generation, and hybrid retrieval generation with a fine-tuned JSON intent router.

The operational boundary is support guidance. The system classifies raw customer queries, retrieves relevant corporate SOP passages, generates grounded response guidance, and flags escalation needs. It does not directly mutate accounts, process refunds, process payments, or override policies.

Expected router JSON:

```
{"intent": "refund_request", "category": "REFUND_POLICY"}
```

Methodology

Stage 1 profiles customer interaction data and SOP documents, including prompt length, intent balance, duplicate detection, and SOP TF-IDF similarity. Stage 2 standardizes records, creates strict ChatML targets, performs normalized deduplication, and creates stratified train/validation/test splits. Stage 3 establishes a deterministic baseline and a naive RAG system using SOP retrieval. Stage 4 fine-tunes a LoRA/QLoRA router that emits strict JSON and uses the extracted intent/category to improve retrieval.

Success Criteria

Router format adherence should be at least 95 percent. Intent accuracy should improve after fine-tuning. SOP-grounded RAG should reduce unsupported policy claims compared with baseline generation. Deterministic inference should produce stable answers across repeated runs.